

1. Write a Remote Object Communication Program to create a Student table and display records.

In Eclipse need to add jar file also

Query	Query History
1	CREATE TABLE student (
2	student_id INTEGER PRIMARY KEY,
3	student_name VARCHAR(100),
4	student_course VARCHAR(100)
5	);

Data Output	Messages	Notifications
CREATE TABLE		
Query returned successfully in 119 msec.		

Query	Query History
1	INSERT INTO student
2	VALUES
3	(1, 'Aditi Sinha', 'MCA'),
4	(2, 'Sejal Patole', 'MCA'),
5	(3, 'Harsh Patil', 'MCA'),
6	(4, 'Suyash Tambade', 'MCA'),
7	(5, 'Manish Patil', 'MCA');

Data Output	Messages	Notifications
INSERT 0 5		
Query returned successfully in 73 msec.		

Query

Query History

1

SELECT * FROM student;

Data Output

Messages

Notifications

SQL

	student_id [PK] integer	student_name character varying (100)	student_course character varying (100)
1	1	Aditi Sinha	MCA
2	2	Sejal Patole	MCA
3	3	Harsh Patil	MCA
4	4	Suyash Tambade	MCA
5	5	Manish Patil	MCA

Student.java

package

```
RMI_JDBC;
import java.io.Serializable;
public class Student implements Serializable {
    private int student_id;
    private String
student_name; private
String student_course;
    public int getStudent_id() {
        return student_id;
    }
    public void setStudent_id(int student_id) {
        this.student_id = student_id;
    }
    public String getStudent_name() {
        return student_name;
    }
    public void setStudent_name(String
student_name) { this.student_name =
student_name;
    }
    public String getStudent_course() {
```

```
    return student_course;
}
public void setStudent_course(String student_course) {
    this.student_course = student_course;
}
}
```

Hello.java (Interface)

```
package RMI_JDBC;
import java.rmi.Remote;
import java.rmi.RemoteException;
import java.util.List;
public interface Hello extends Remote {
    List<Student> getStudents() throws RemoteException, Exception;
    void printMsg() throws RemoteException;
}
```

Server.java

```
package
RMI_JDBC;
import java.rmi.Naming;
import java.rmi.registry.LocateRegistry;
public class Server {
    public static void main(String args[])
    { try {
        LocateRegistry.createRegistry(9000
        ); Hello obj = new ImplExample();
        Naming.rebind( "rmi://localhost:9000/Hello", obj
        );
        System.out.println("RMI Server Started");
    }
    catch (Exception e) {
        e.printStackTrace();
    }
}
```

Client.java

```
package
RMI_JDBC;
import
java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;
import java.util.List;
public class Client {
    public static void main(String args[])
```

```
{ try {  
    Registry registry = LocateRegistry.getRegistry("localhost",  
    9000); Hello stub = (Hello) registry.lookup("Hello");  
    List<Student> list = stub.getStudents();  
    for (Student s : list) {  
        System.out.println("Aditi Sinha B-58");  
        System.out.println("Student ID: " + s.getStudent_id());  
    }  
}
```

```
        System.out.println("Student Name : "+
        s.getStudent_name()); System.out.println("Course : "+
        s.getStudent_course()); System.out.println(" ");
    }
} catch (Exception e) {
    System.out.println("Client Exception : " + e);
    e.printStackTrace();
}
}
}
```

ImplExample.java

package

RMI_JDBC;

import java.rmi.RemoteException;

import java.rmi.server.UnicastRemoteObject;

import java.sql.Connection;

import

java.sql.DriverManager;

import java.sql.ResultSet;

import java.sql.Statement;

import java.util.ArrayList;

import java.util.List;

public class ImplExample **extends** UnicastRemoteObject **implements** Hello {

protected ImplExample() **throws** RemoteException {

super();

}

@Override

public List<Student> getStudents() **throws** Exception {

List<Student> list = **new** ArrayList<>();

Class.forName("org.postgresql.Driver");

String DB_URL = "jdbc:postgresql://localhost:5432/library";

String USER = "postgres";

String PASS = "studentAditi";

Connection conn = DriverManager.getConnection(DB_URL, USER, PASS);

System.out.println("Database Connected Successfully");

Statement stmt =

conn.createStatement(); String sql =

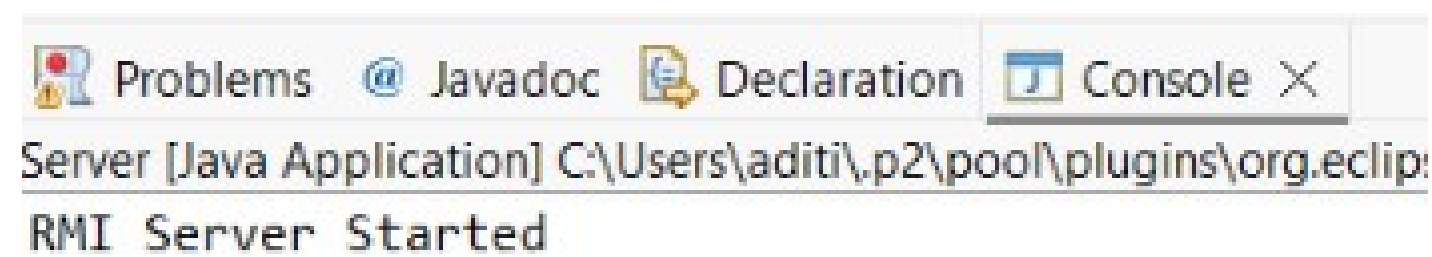
"SELECT * FROM student"; ResultSet rs

= stmt.executeQuery(sql); **while**

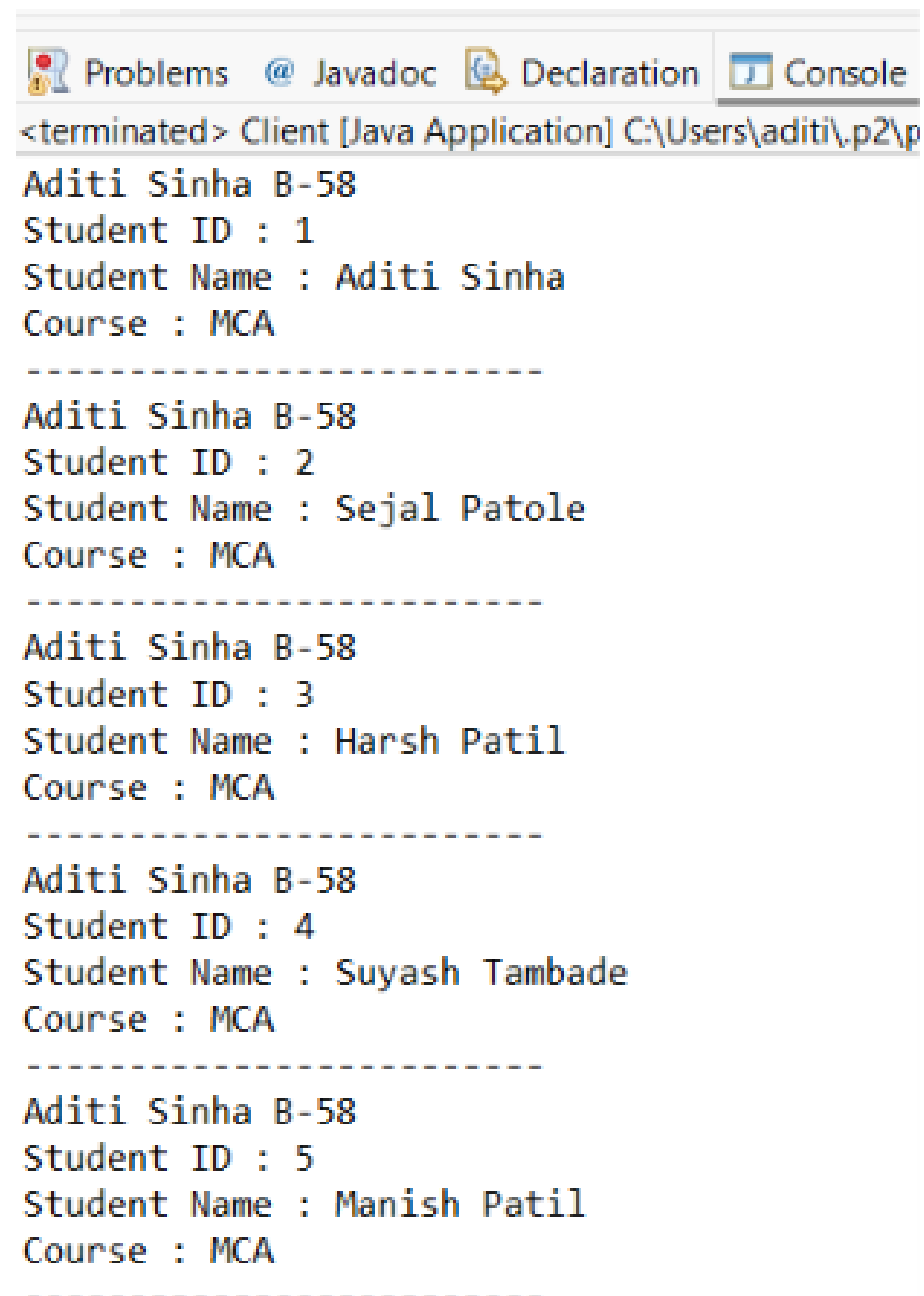
(rs.next()) {

```
Student s = new Student();  
s.setStudent_id(rs.getInt("student_id"));  
s.setStudent_name(rs.getString("student_name"));  
s.setStudent_course(rs.getString("student_course"));  
list.add(s);  
}
```

```
rs.close();
stmt.close();
conn.close();
return list;
}
@Override
public void printMsg() throws RemoteException {
    System.out.println("Student RMI Service Working");
}
}
```

Output:

Problems Javadoc Declaration Console X
Server [Java Application] C:\Users\aditi\p2\pool\plugins\org.eclipse
RMI Server Started



Problems Javadoc Declaration Console
<terminated> Client [Java Application] C:\Users\aditi\p2\p
Aditi Sinha B-58
Student ID : 1
Student Name : Aditi Sinha
Course : MCA

Aditi Sinha B-58
Student ID : 2
Student Name : Sejal Patole
Course : MCA

Aditi Sinha B-58
Student ID : 3
Student Name : Harsh Patil
Course : MCA

Aditi Sinha B-58
Student ID : 4
Student Name : Suyash Tambade
Course : MCA

Aditi Sinha B-58
Student ID : 5
Student Name : Manish Patil
Course : MCA

2. Write a Remote Object Communication Program to create a electricity Bill table and display records.

Query	Query History
1 2 3 4 5 6	<pre>CREATE TABLE electricity_bill (bill_id INTEGER PRIMARY KEY, customer_name VARCHAR(100), units INTEGER, amount DECIMAL(10,2));</pre>
Data Output	Messages
CREATE TABLE	Query returned successfully in 92 msec.

Query	Query History
1 2 3 4 5 6 7	<pre>INSERT INTO electricity_bill VALUES (1, 'Aditi Sinha', 120, 850.50), (2, 'Sejal Patole', 250, 1750.75), (3, 'Manish Patil', 180, 1200.00), (4, 'Suyash Tambade', 300, 2200.50), (5, 'Harsh Patil', 300, 2200.50);</pre>
Data Output	Messages
INSERT 0 5	Query returned successfully in 64 msec.

Query

Query History

1

SELECT * FROM electricity_bill;

Data Output

Messages

Notifications

SQL

	bill_id [PK] integer	customer_name character varying (100)	units integer	amount numeric (10,2)
1	1	Aditi Sinha	120	850.50
2	2	Sejal Patole	250	1750.75
3	3	Manish Patil	180	1200.00
4	4	Suyash Tambade	300	2200.50
5	5	Harsh Patil	300	2200.50

Electricity.java

```
package RMI_JDBC;

import java.io.Serializable;

public class ElectricityBill implements Serializable {
    private int bill_id;
    private String customer_name;
    private int units;
    private double amount;
    public int getBill_id() {
        return bill_id;
    }
    public void setBill_id(int bill_id) {
        this.bill_id = bill_id;
    }
    public String getCustomer_name() {
        return customer_name;
    }
    public void setCustomer_name(String customer_name) {
        this.customer_name = customer_name;
    }
    public int getUnits() {
        return units;
    }
    public void setUnits(int units) {
        this.units = units;
    }
    public double getAmount() {
```

Name Aishwarya Pedgaonkar

Roll No 45 .

Div B2

```
    return amount;  
}  
public void setAmount(double amount) {
```

```
    this.amount = amount;
}
}
```

Hello.java

```
package RMI_JDBC;
import java.rmi.Remote;
import java.rmi.RemoteException;
import java.util.List;
public interface Hello extends Remote {
    List<ElectricityBill> getBills()
        throws RemoteException, Exception;
    void printMsg() throws RemoteException;
}
```

Client.java

```
package RMI_JDBC;
import
java.rmi.registry.LocateRegistry;
import java.rmi.registry.Registry;
import java.util.List;
public class Client {
    public static void main(String args[]) {
        try {
            Registry registry =
                LocateRegistry.getRegistry("localhost",9000); Hello stub =
                (Hello) registry.lookup("Hello"); List<ElectricityBill> list =
                stub.getBills();
            for (ElectricityBill b : list) {
                System.out.println("Aditi Sinha B-58");
                System.out.println("Bill ID : "+ b.getBill_id());
                System.out.println("Customer Name : "+
                    b.getCustomer_name()); System.out.println("Units : "+
                    b.getUnits()); System.out.println("Amount : ₹ "+
                    b.getAmount()); System.out.println("- ");
            }
        } catch (Exception e) {
            System.out.println("Client Exception : " + e);
            e.printStackTrace();
        }
    }
}
```

Server.java

```
package RMI_JDBC;
import java.rmi.Naming;
```

Name Aishwarya Pedgaonkar

Roll No 45 .

Div B2

```
import java.rmi.registry LocateRegistry;  
public class Server {  
    public static void main(String args[]) {  
        try {  
            LocateRegistry.createRegistry(9000);
```

```
        Hello obj = new ImplExample();
        Naming.rebind("rmi://localhost:9000/Hello", obj);
        System.out.println("RMI Server Started");
    }
    catch (Exception e) {
        e.printStackTrace();
    }
}
}
```

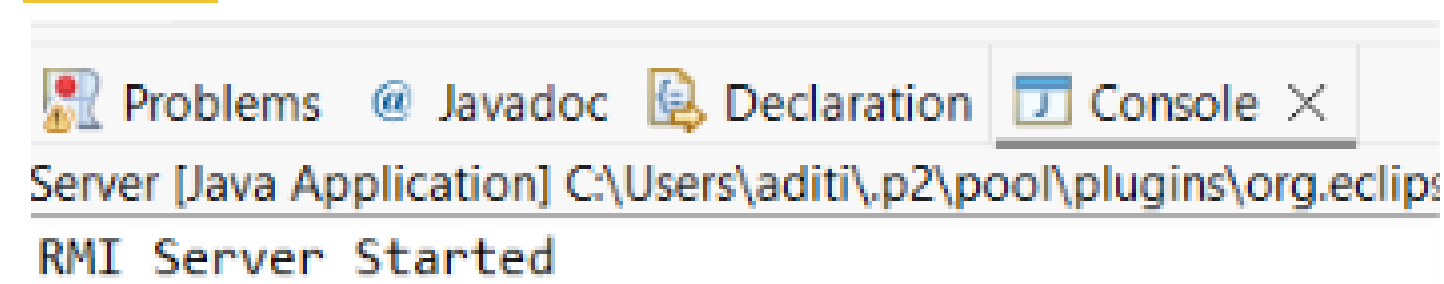
ImplExample.java

```
package RMI_JDBC;
import java.rmi.RemoteException;
import java.rmi.server.UnicastRemoteObject;
import java.sql.Connection;
import
java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
import java.util.ArrayList;
import java.util.List;
public class ImplExample extends UnicastRemoteObject
    implements Hello {
    protected ImplExample() throws RemoteException {
        super();
    }
    @Override
    public List<ElectricityBill> getBills() throws Exception {
        List<ElectricityBill> list = new ArrayList<>();
        Class.forName("org.postgresql.Driver");
        String DB_URL = "jdbc:postgresql://localhost:5432/library";
        String USER = "postgres";
        String PASS = "studentAditi";
        Connection conn =
        DriverManager.getConnection(DB_URL,USER,PASS);
        System.out.println("Database Connected Successfully");
        Statement stmt
        =conn.createStatement(); String sql =
        "SELECT * FROM electricity_bill"; ResultSet
        rs = stmt.executeQuery(sql);
        while (rs.next()) {
            ElectricityBill b = new ElectricityBill();
            b.setBill_id(rs.getInt("bill_id"));
            b.setCustomer_name(rs.getString("customer_name"
            )); b.setUnits(rs.getInt("units"));
        }
    }
}
```

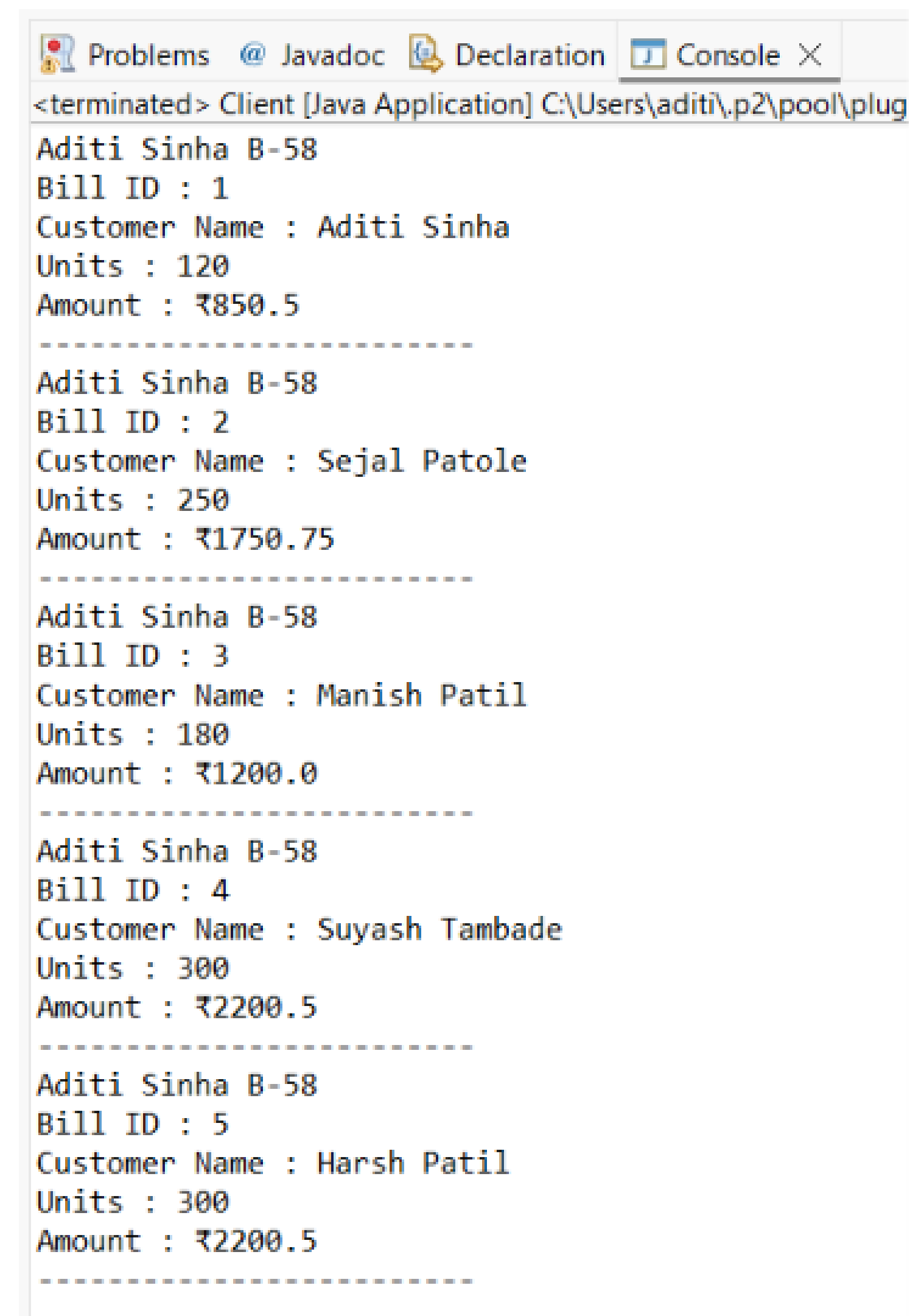
```
        b.setAmount(rs.getDouble("amount"));
        list.add(b);
    }
    rs.close();
    stmt.close();
    conn.close();
    return list;
```

```
}  
@Override  
public void printMsg()  
    throws  
        RemoteException{  
    System.out.println("Electricity Bill RMI Service Working");  
}  
}
```

Output



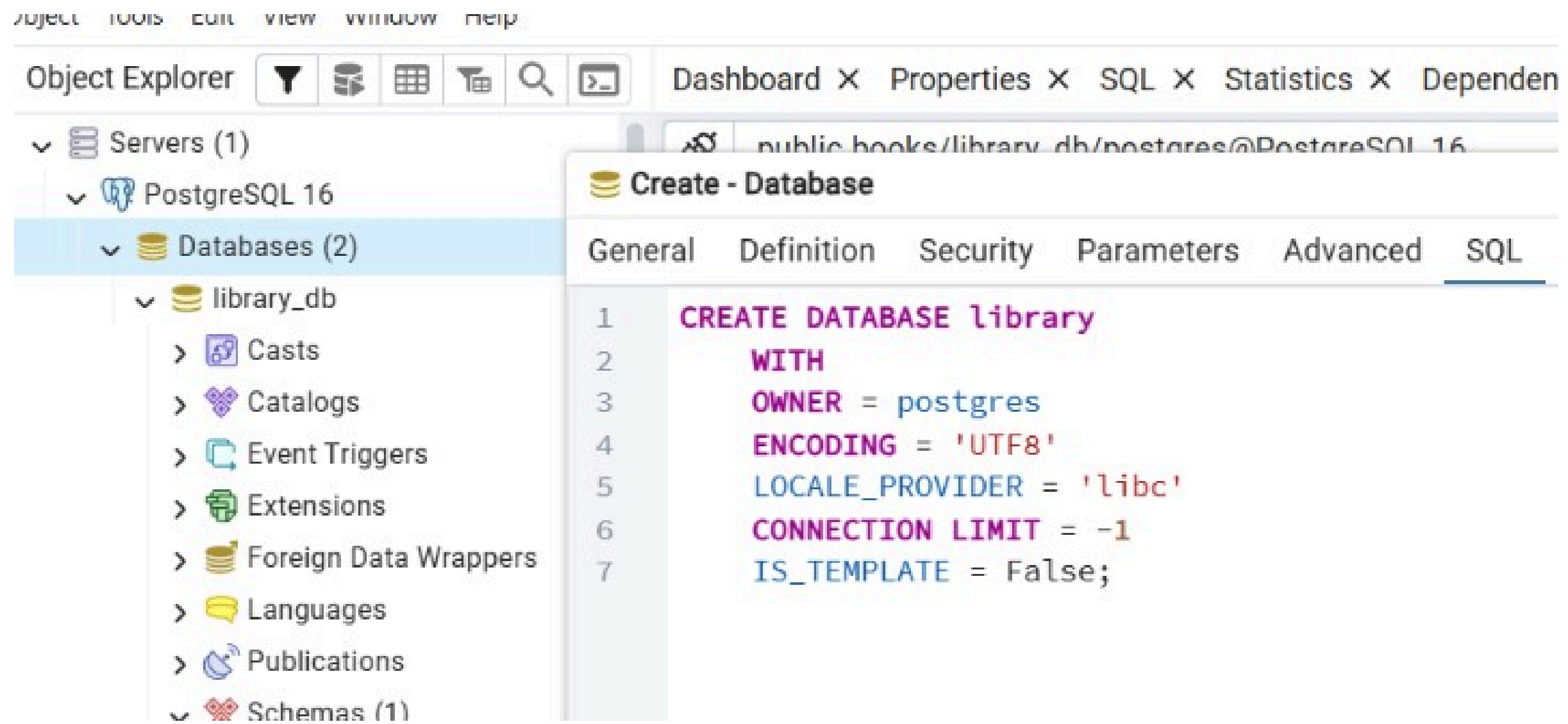
The screenshot shows the Eclipse IDE's Console window. The title bar includes 'Problems', 'Javadoc', 'Declaration', and 'Console'. The console output displays the message 'RMI Server Started'.

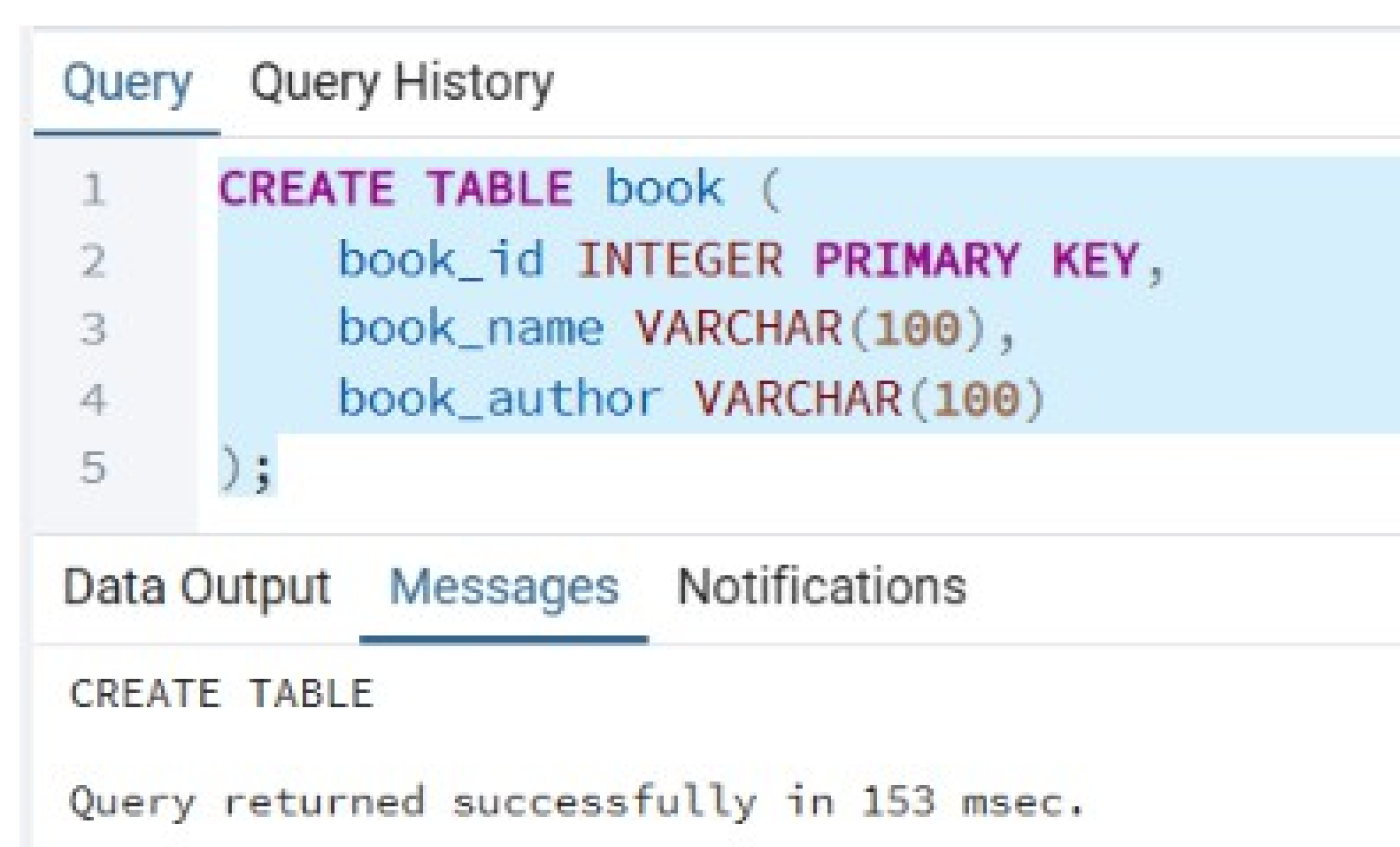
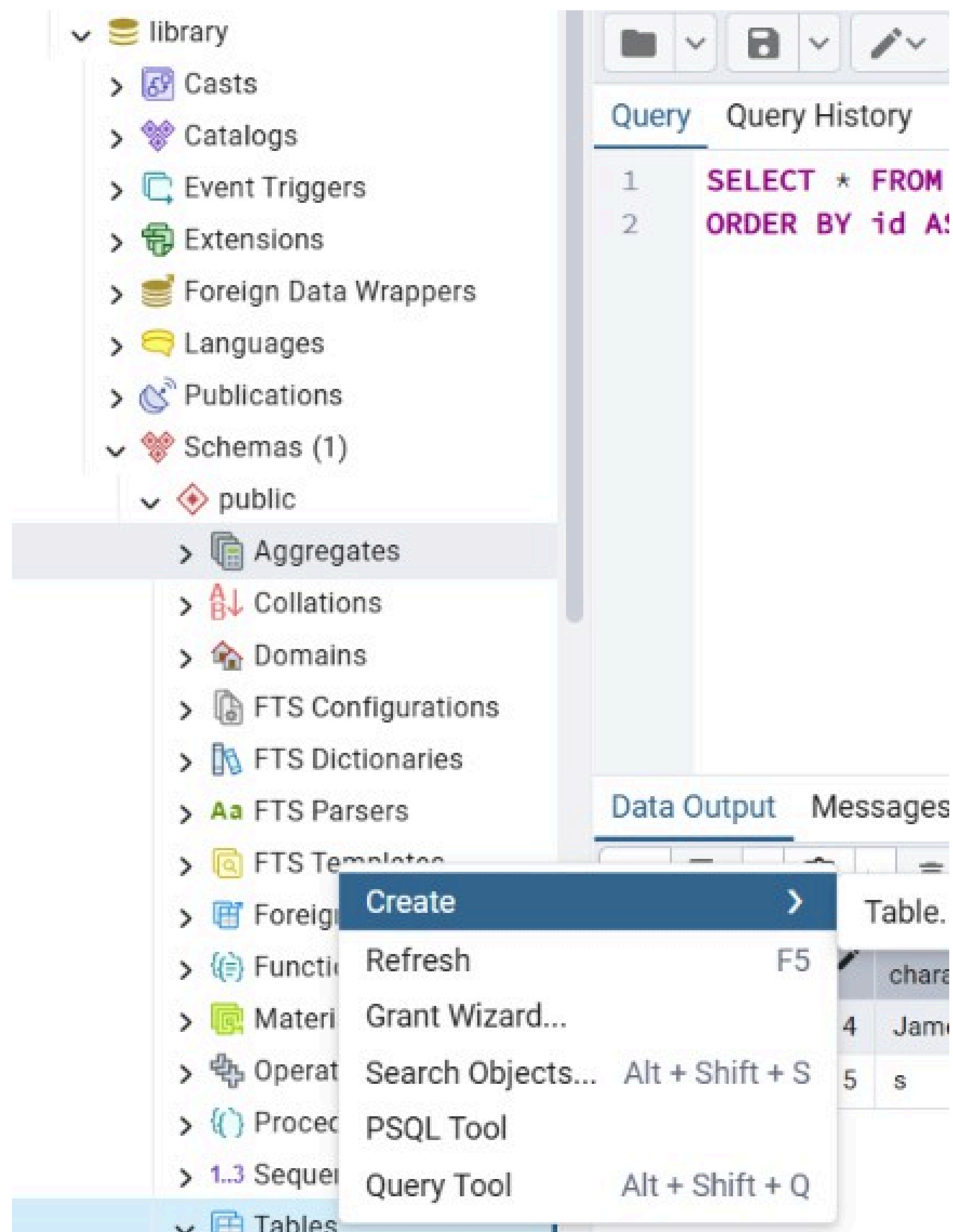


The screenshot shows the Eclipse IDE's Console window with the title bar 'Problems', 'Javadoc', 'Declaration', and 'Console'. The console output shows the results of a client application, displaying five bills with their details separated by dashed lines.

```
<terminated> Client [Java Application] C:\Users\aditi\p2\pool\plug  
Aditi Sinha B-58  
Bill ID : 1  
Customer Name : Aditi Sinha  
Units : 120  
Amount : ₹850.5  
-----  
Aditi Sinha B-58  
Bill ID : 2  
Customer Name : Sejal Patole  
Units : 250  
Amount : ₹1750.75  
-----  
Aditi Sinha B-58  
Bill ID : 3  
Customer Name : Manish Patil  
Units : 180  
Amount : ₹1200.0  
-----  
Aditi Sinha B-58  
Bill ID : 4  
Customer Name : Suyash Tambade  
Units : 300  
Amount : ₹2200.5  
-----  
Aditi Sinha B-58  
Bill ID : 5  
Customer Name : Harsh Patil  
Units : 300  
Amount : ₹2200.5  
-----
```

3. Write a Remote Object Communication Program to create a Book table and display records.





Query

Query History

1

INSERT INTO book VALUES

2

(1, 'Java Programming', 'Herbert Schildt'),

3

(2, 'Database System Concepts', 'Korth'),

4

(3, 'Computer Networks', 'Andrew Tanenbaum');

Data Output

Messages

Notifications

INSERT 0 3

Query returned successfully in 142 msec.

Query

Query History

1

SELECT * FROM book;

Data Output

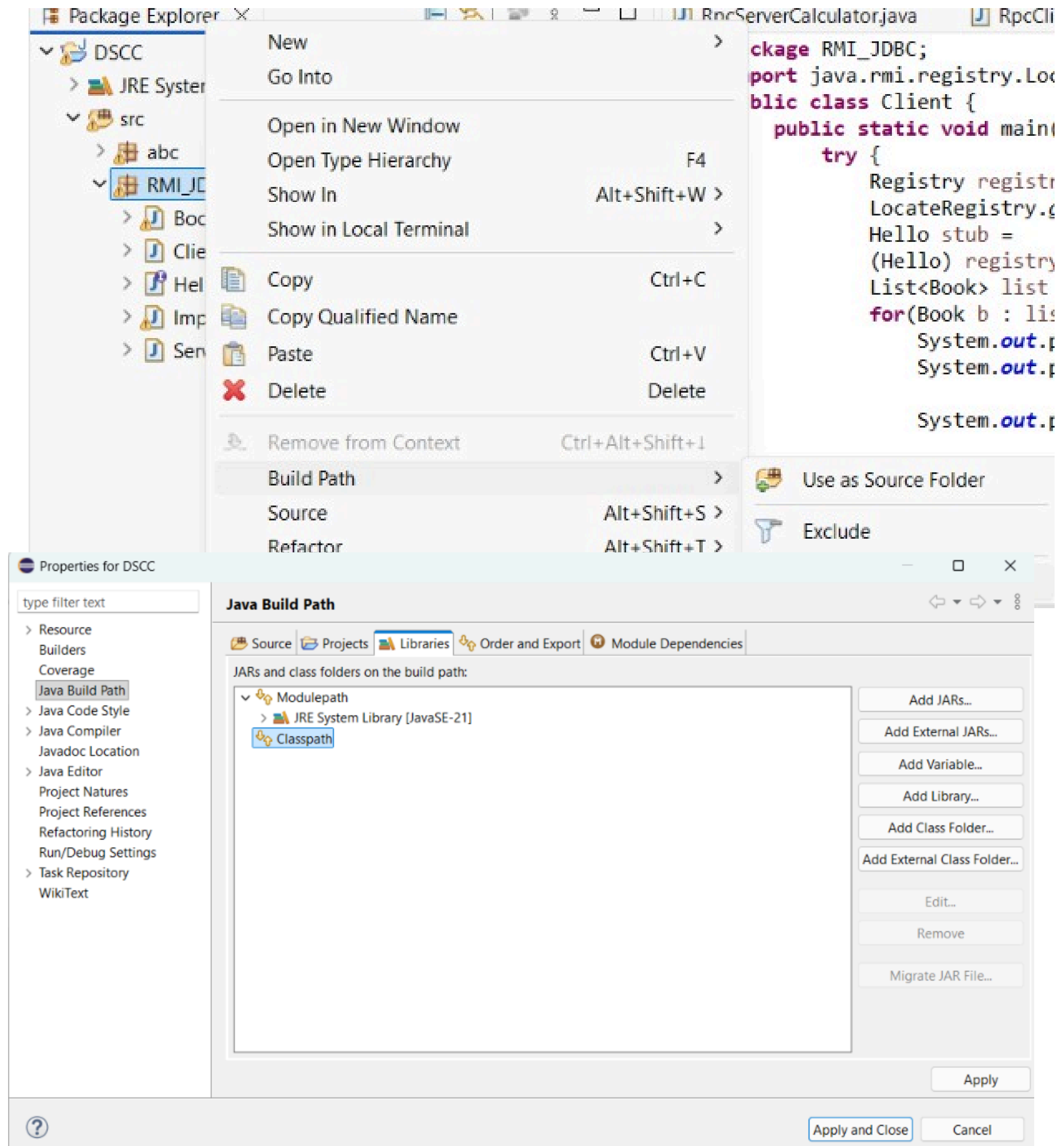
Messages

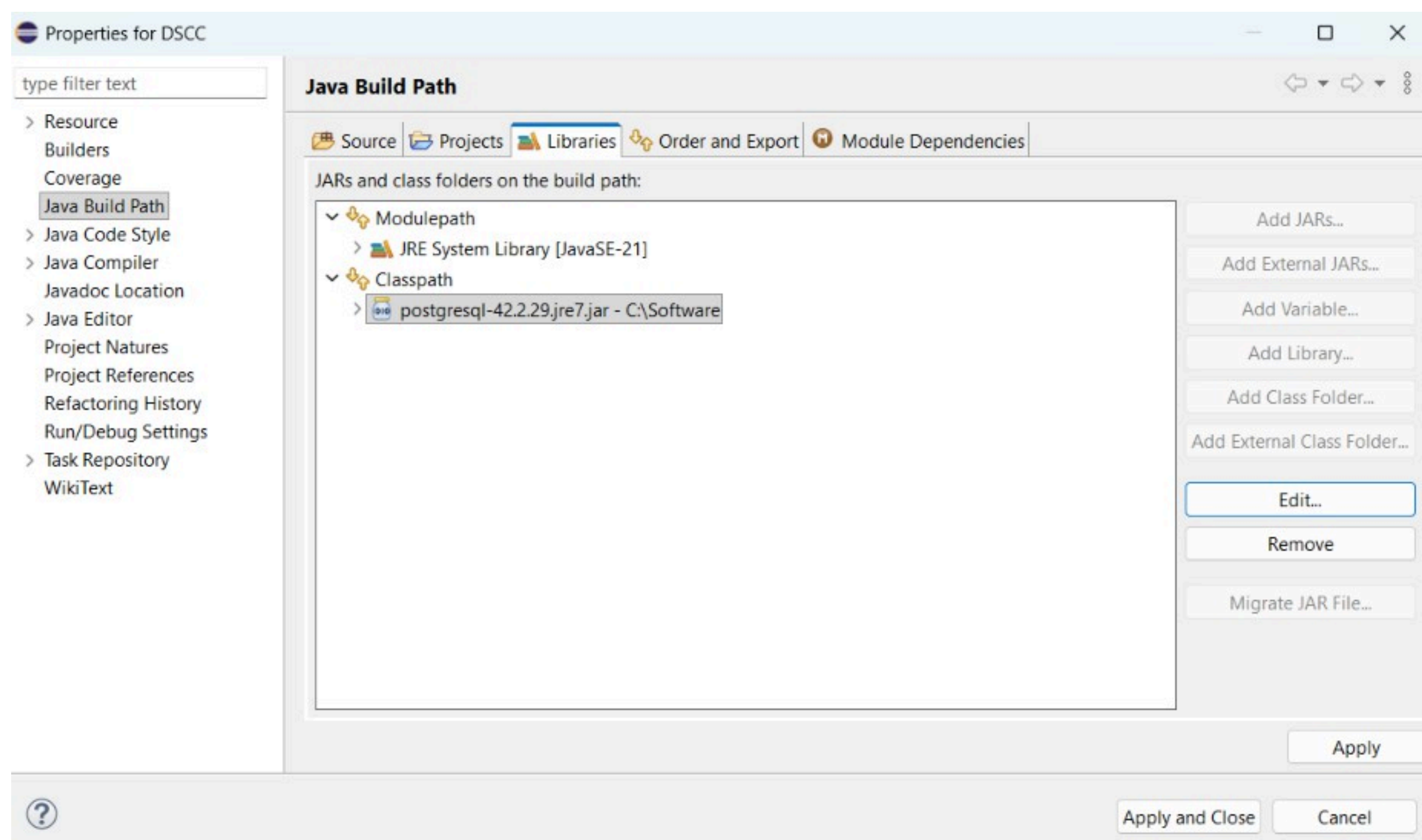
Notifications

≡+

SQL

	book_id [PK] integer	book_name character varying (100)	book_author character varying (100)
1	1	Java Programming	Herbert Schildt
2	2	Database System Conce...	Korth
3	3	Computer Networks	Andrew Tanenbaum





Book.java

```
package roc;
import java.io.Serializable;
public class Book implements Serializable {
    private int book_id;
    private String book_name;
    private String book_author;
    public int getBook_id() {
        return book_id;
    }
    public void setBook_id(int book_id) {
        this.book_id = book_id;
    }
    public String getBook_name()
    { return book_name;
    }
    public void setBook_name(String book_name) {
        this.book_name = book_name;
    }
    public String getBook_author() {
        return book_author;
    }
}
```

```
}  
public void setBook_author(String book_author) {  
    this.book_author = book_author;  
}
```

```
}
```

Hello.java —interface

```
package roc;
```

```
import java.rmi.Remote;
```

```
import
```

```
java.rmi.RemoteException;
```

```
import java.util.List;
```

```
public interface Hello extends Remote {
```

```
    List<Book> getBooks() throws RemoteException, Exception;
```

```
    void printMsg() throws RemoteException;
```

```
}
```

Client.java

```
package roc;
```

```
import
```

```
java.rmi.registry.LocateRegistry;
```

```
import java.rmi.registry.Registry;
```

```
import java.util.List;
```

```
public class Client {
```

```
    public static void main(String args[]) {
```

```
        try {
```

```
            Registry registry =
```

```
                LocateRegistry.getRegistry("localhost",9000);
```

```
Hello stub =  
(Hello) registry.lookup("Hello");
```

```
List<Book> list = stub.getBooks();
```

```

for(Book b : list) { System.out.println(
    "Aditi Sinha B-58" );
    System.out.println("ID : " + b.getBook_id());

    System.out.println("Name : " + b.getBook_name());

    System.out.println("Author : " + b.getBook_author());

    System.out.println("----- ");
}

}
catch(Exception e) {

    System.out.println("Client Exception : " +

    e);

    e.printStackTrace();
}
}
}

```

Server.java

```

package roc;
import java.rmi.Naming;
import java.rmi.registry.LocateRegistry;

public class Server {
    public static void main(String args[]) {
        try {
            LocateRegistry.createRegistry(900
0); Hello obj = new ImplExample();
            Naming.rebind( "rmi://localhost:9000/Hello", obj
            );
            System.out.println("RMI Server Started");
        }
        catch(Exception e) {

```

```
        e.printStackTrace();
    }
}
```

ImplExample.java

```
package roc;
import java.rmi.RemoteException;
import
java.rmi.server.UnicastRemoteObject;
import java.sql.Connection;
import
java.sql.DriverManager;
import java.sql.ResultSet;
import java.sql.Statement;
import java.util.ArrayList;
import java.util.List;

public class ImplExample extends UnicastRemoteObject implements Hello {
    protected ImplExample() throws RemoteException {
        super();
    }
    @Override
    public List<Book> getBooks() throws Exception {
        List<Book> list = new ArrayList<>();

        // PostgreSQL Driver
        Class.forName("org.postgresql.Driver"
        );

        // PostgreSQL Database Details
        String DB_URL = "jdbc:postgresql: //localhost:5432/library";
        String USER = "postgres";
        String PASS = "password";           // Change this to your pgAdmin password

        // Database Connection
        Connection conn = DriverManager.getConnection(DB_URL, USER, PASS);
        System.out.println("Database Connected Successfully by Aditi-B58" );
        Statement stmt = conn.createStatement();

        // Table name created in pgAdmin
        String sql = "SELECT * FROM book";
        ResultSet rs =
        stmt.executeQuery(sql);
        while(rs.next()) {
```

```
Book b = new Book();
b.setBook_id(rs.getInt("book_id"));
b.setBook_name(rs.getString("book_name"))
;
b.setBook_author(rs.getString("book_author"
)); list.add(b);
}
rs.close();
stmt.close();
```

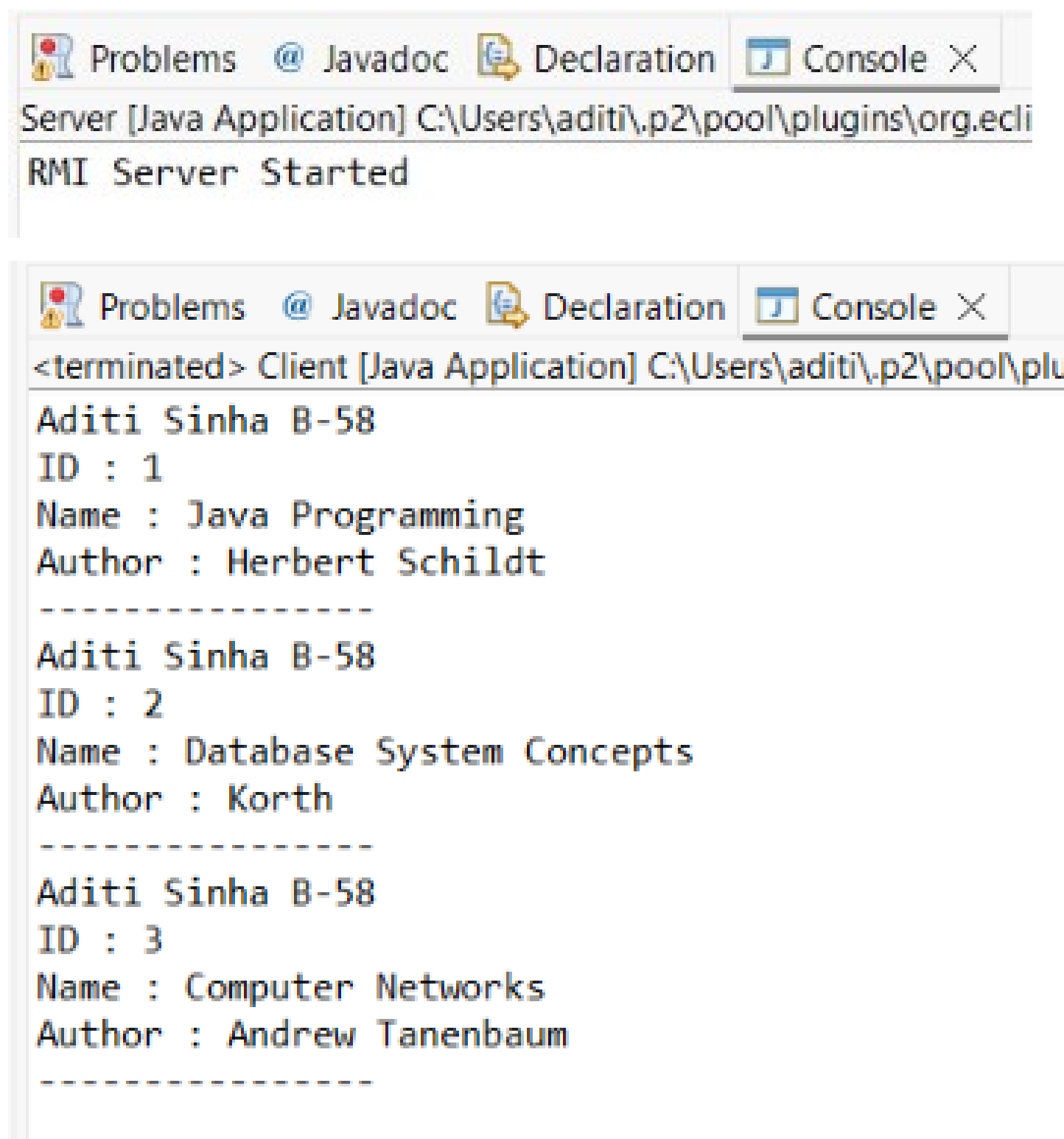
```

        conn.close();
        return list;
    }

    @Override
    public void printMsg() throws RemoteException {
        System.out.println("Library RMI Service Working");
    }
}

```

Output:



The screenshot shows two separate Eclipse IDE console windows. The top window, titled 'Server [Java Application]', displays the message 'RMI Server Started'. The bottom window, titled '<terminated> Client [Java Application]', shows the output of a client application. It lists three books with their IDs, names, and authors, separated by dashed lines. The books are: ID 1, 'Java Programming' by Herbert Schildt; ID 2, 'Database System Concepts' by Korth; and ID 3, 'Computer Networks' by Andrew Tanenbaum.

```

Server [Java Application] C:\Users\aditi\p2\pool\plugins\org.ecl
RMI Server Started

<terminated> Client [Java Application] C:\Users\aditi\p2\pool\plu
Aditi Sinha B-58
ID : 1
Name : Java Programming
Author : Herbert Schildt
-----
Aditi Sinha B-58
ID : 2
Name : Database System Concepts
Author : Korth
-----
Aditi Sinha B-58
ID : 3
Name : Computer Networks
Author : Andrew Tanenbaum
-----

```